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Mitigating Fracture Height Growth: Fracturing Design Sets Strategy in Field Development

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Abstract

A field in northern Oman within the largest Block X (block number remains confidential), is characterized by heterogeneous sandstone lithology with a relatively thin gross reservoir, confined between two distinct shale formations. Diagnostic Fracture Injection Tests (DFITs) indicated significant fracture height growth potential. This paper presents an engineered hydraulic fracturing strategy, integrating real-time data and field trials, to mitigate downward fracture propagation and optimize water injector performance, enhancing reservoir pressure support within the developing field.

A conventional hydraulic fracturing treatment, modeled in a 3D simulator, was initially designed to enhance reservoir conductivity and injectivity. However, the lower shale formation posed a significant risk of excessive fracture height growth and compromising the fracture conductivity. To mitigate this risk, the initial fracturing design incorporated classic containment strategies, including lowered to 20% pad volume and a reduced pumping rate of 2.25 m³/min. However, based on pre-job DFIT and G-function analysis confirmed height recession, with net pressure values as low as 3.9 MPa, indicating the potential for uncontrolled height growth. To address this, a controlled proppant bridging barrier was strategically placed during the fluid efficiency test (FET), based on SPE literature and field learnings. This approach effectively limited fracture growth into the lower formation.

The primary objective of placing 40 metric tons (MT) of proppant was successfully achieved. Post-treatment tracer logs confirmed controlled fracture height growth, ensuring the integrity of the target zone. The well's injectivity improved significantly, with post-fracturing performance exceeding expectations by a factor of 14. This enhancement helped maintain pressure in nearby oil producers, resulting in a 10% increase in daily production. The operation validated the importance of adaptive fracturing design and real-time diagnostics in overcoming containment challenges. The revised strategy demonstrated a reliable, field-proven method for managing fracture geometry in thin, heterogeneous sandstone reservoirs typical of Project Field and similar fields.

This study introduces a novel integration of real-time diagnostic data with engineered proppant bridging to manage fracture height in challenging stratigraphy. The approach mitigated downward fracture growth, safeguarding zonal isolation and enhancing injector performance in Project Field. These findings offer

valuable insights for operators managing height containment risks in analogous reservoirs and contribute to the evolution of optimized fracturing practices in northern Oman.

Introduction

The discussed well is named a "Project Well" in this publication, targeting the "Project Formation" of the "Project Field", represents a strategic effort to enhance injectivity by repurposing a former ESP-supported oil producer into a water injector. As part of this conversion, a carefully engineered hydraulic fracturing treatment was designed and executed to establish a high-conductivity flow path within a vertically constrained, moderately depleted sandstone interval.

The Project Formation reservoir offers ~7 meters of net pay within a gross interval of ~31 meters, with lithological boundaries formed by interbedded siltstone and claystone. While porosity (~12%) and permeability (~5 mD) are favorable, the potential for uncontrolled vertical fracture propagation posed a critical risk. Without effective height containment, the fracture could extend into non-reservoir zones, dilute proppant distribution, reduce net pressure buildup, and compromise conductivity — particularly if the bottom intersects lower-stress or incompetent rock.

In such conditions, confidence in the fracture growth model becomes central to treatment design. In the Project Well, Diagnostic Fracture Injection Testing (DFIT) revealed a pressure decline profile that aligned closely with the geomechanical model derived in a 3D simulator, providing a valuable calibration for stress and barrier integrity.

The fracturing treatment was preceded by a controlled proppant bridging barrier — a pre-pad injection stage containing a designed blend of coarse and fine solids — specifically formulated to mitigate downward fracture propagation. During main stage pumping, pressure trends indicated consistent net pressure response, with no loss of confinement beyond the perforated interval. Post-fracture diagnostics, including temperature and tracer monitoring, further confirmed that fracture growth remained confined within the target layer.

This paper presents a stepwise evaluation of the well, covering reservoir characterization, stress modeling, stimulation design, real-time observations, and post-job interpretation. The Project well case demonstrates how height growth control — when validated by diagnostics and modeled accurately — becomes a cornerstone of effective fracture design in thin or stress-sensitive sandstone injectors.

Field Summary and Reservoir Characterization

The Project Well lies in the western portion of the Project Field, within license Block X — a vast hydrocarbon contract area in northern Oman (~90,000 km²), extending westward toward the Saudi Arabian border. Structural and stratigraphic trends in this region link the Project location to major operator's accumulations within Block X ([EDO Investor Presentation, 2025](#); [Wood Mackenzie, 2024](#)).

The target reservoir is the Jurassic-aged Project Formation, a coastal to shallow-marine sandstone deposited during the Middle Jurassic. While Jurassic formations are typically deep in the Arabian Peninsula, here the unit was encountered at a shallow depth (~1000–1150 mMD). This is attributed to regional uplift and mild inversion of Jurassic normal faults during the Late Cretaceous — a structural style also documented in the Project Field area ([Bazalgette & Salem, 2019](#); [Filbrandt et al., 2006](#)).

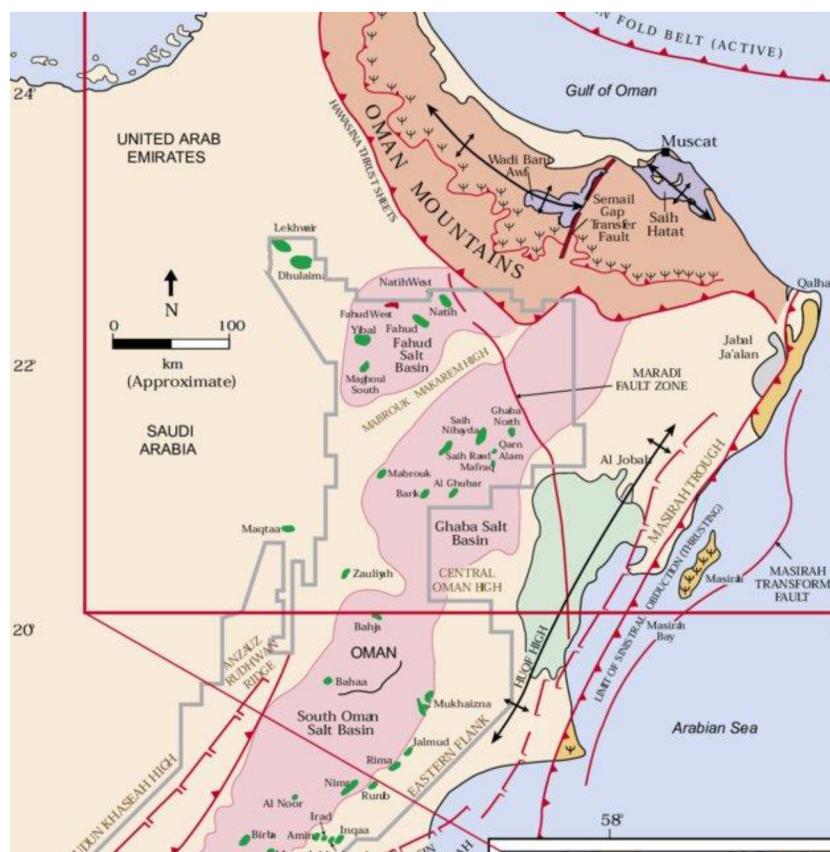


Figure 1—Project Field Location in the North-East flank of the Block X

The treated interval comprises ~7 meters of net clean sandstone, bounded above and below by transitional shale and claystone. These bounding layers provide limited mechanical contrast, with no distinct sealing properties, increasing the risk of vertical fracture growth, particularly downward into less competent formations.

Reservoir characteristics include:

- Effective porosity: ~12%
- Permeability: 1–64 mD (average ~5 mD)
- Reservoir pressure: ~10 MPa (slight depletion)
- Temperature: ~66°C
- Fluid history: light oil producer (39° API, ~0.65 cP viscosity)

A vertical stress profile was constructed using log-derived mechanical properties and modeled in a 3D simulator. This model was validated against offset well data and refined through pre-frac DFIT interpretation, which confirmed minimal stress contrast below the perforated interval. The absence of a cement bond log was offset by openhole logs and formation top analysis, which similarly suggested no significant mechanical barrier above or below the pay zone.

To assess mechanical layering, neutron-density and gamma ray-derived shale volumes were compared, with the neutron-density curve showing marginally better trends for identifying weak barrier intervals — and was adopted in final modeling.

Given the thin pay, weak bounding layers, and lack of natural fractures or strong anisotropy, the design approach prioritized containment over extension. This was achieved through conservative fracture

modeling, supplemented by pre-frac diagnostics and the placement of a controlled proppant bridging barrier to reinforce vertical control prior to the main treatment.

The Project Well's unique setting — a shallow Jurassic target within the uplifted western Block X margin — imposed strict demands on vertical control. These geological and mechanical realities shaped a treatment strategy that favored containment, diagnostics, and barrier placement over stimulation intensity.

Fracturing Fluid and Proppant Design

In the Project Well, fluid and proppant design considerations were dictated by operational practicality and compatibility with the well's depth, pressure regime, and equipment constraints, rather than by extreme formation demands. The reservoir temperature ($\sim 66^{\circ}\text{C}$) and moderate bottomhole pressure (~ 10 MPa) allowed for the use of conventional gel systems without significant thermal or chemical modifications.

A delayed crosslinked guar-based fluid system (25 lb gel loading) was selected primarily for its reliability in low-rate operations and its proven performance in similar sandstone environments. The system provided stable viscosity during pumping, controlled leakoff, and was supported by a predictable breaker package that allowed timely gel cleanup after the treatment.

From a proppant standpoint, the treatment used intermediate strength proppants (ISP) in 30/50 and 20/40 mesh for the main stages, with 16/30 ISP tail-in to reinforce near-wellbore conductivity. The maximum concentration was capped at 7 PPA, a value balanced between equipment limitations and the stress capacity of the Project Formation.

Conductivity expectations were based on manufacturer data, and a modest proppant overage ($\sim 10\%$) was provisioned to support potential on-the-fly changes during the job. The design objective was not to push proppant intensity or test the fluid's carrying limits, but to deliver a consistent and controllable placement that would support injectivity without risking screenout or excessive pressure escalation.

Fracture Design and Simulation

The primary design challenge in the Project Well was not maximizing fracture half-length or proppant mass, but rather ensuring that the fracture remained geometrically confined within the targeted Project Formation interval. At only 7 meters of net pay and with limited stress contrast across adjacent layers, any unrestrained downward growth could significantly dilute stimulation effectiveness and compromise injectivity.

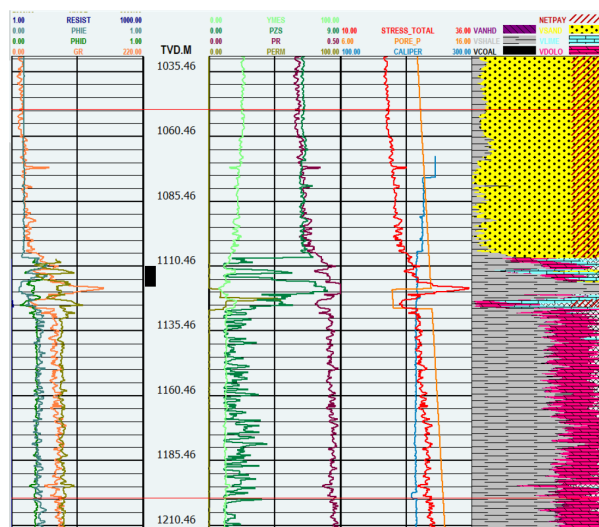


Figure 2—Reservoir Grid Model. Perforation: 1,110.6 mMD to 1,117.60 mMD indicated in black rectangle.

To assess this risk and refine the job envelope, a 3D simulator was used to study reservoir data log and simulated (in 3D software) reservoir grid. The multi-layer model included:

- Young's modulus and Poisson's ratio profiles derived from well logs
- Layer-specific porosity and permeability variations
- Minimum horizontal stress estimates from log-derived mechanical contrasts
- Pore pressure estimated at ~10.5 MPa under minor depletion

Baseline simulations using an 80 MT, 3.0 m³/min design indicated excessive fracture height development — extending beyond 30 meters and breaching lower boundaries. In response, the design was progressively downsized.

The final selected case — 60 MT at 2.25 m³/min — predicted:

- Fracture height: ~21 meters (extending below the perforated interval)
- Fracture half-length: ~45–55 meters
- Maximum net pressure: ~2.5 MPa during main slurry
- Fracture width: sufficient to support 7 PPA using intermediate strength proppant

While the simulated fracture height exceeded the 7-meter perforated zone, the simulator indicated that the dominant fracture aperture and conductivity remained within the Project Formation sandstone, with reduced aperture in the lower wings due to stress and modulus gradients.

Given the height sensitivity, a geometry control mechanism was introduced. A **controlled proppant bridging barrier** was strategically placed prior to the main fracture treatment. This was executed as a separate stage between the calibration injection and proppant slurry, using a blend of coarse and fine solids suspended in viscous gel. The objective was to intentionally bridge at or near the fracture lower tip, thereby increasing near-tip pressure and modifying local growth dynamics.

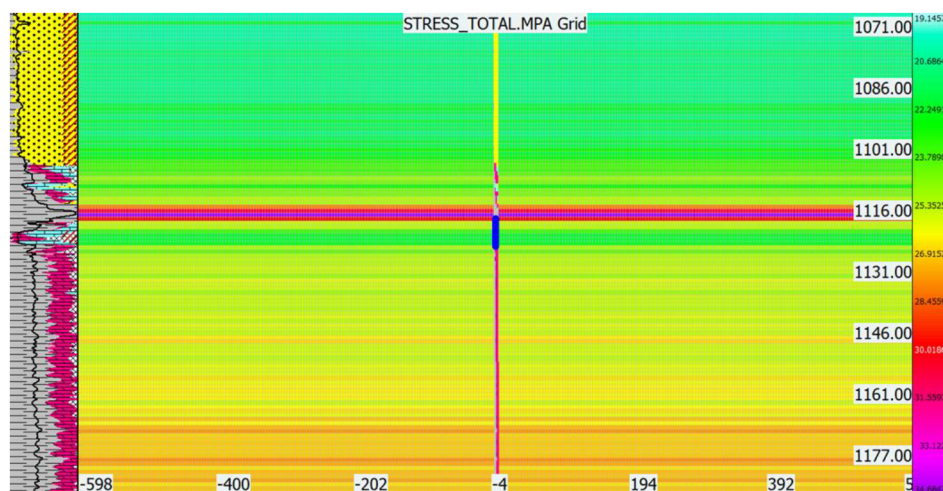


Figure 3—Total Stress illustration resulted from 3D Simulator. Blue vertical bar corresponds to perforation interval. The field with green color (across the perforation interval) corresponds to pay zone stress values. The low-stress barrier below likely to allow fracture propagation downward.

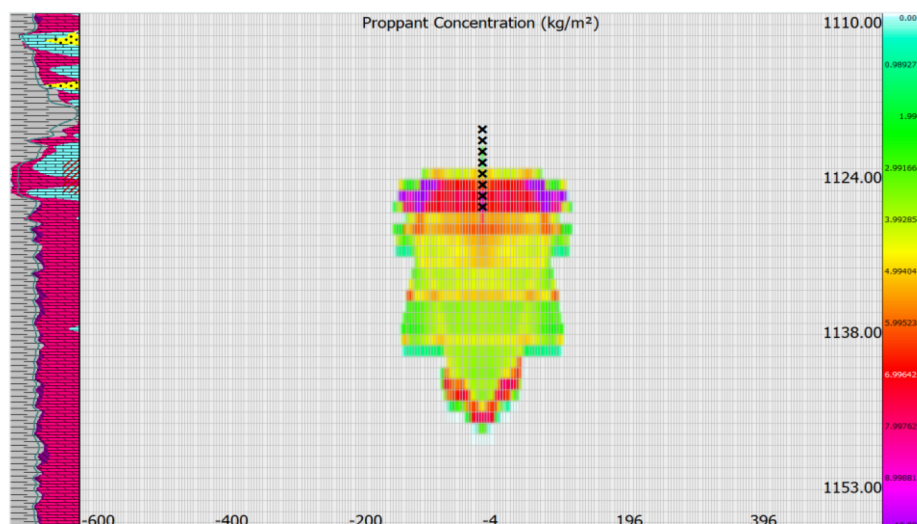


Figure 4—Simulation of Fracture propagation in accord to reservoir grid and treatment design.

Execution Strategy and Pre-Frac Diagnostics

Successful execution of the Project well stimulation hinged on validating the modeled fracture geometry in real time and mitigating the risk of downward growth through proactive diagnostics. Given the case's (Project Well) sensitivity to height propagation and the absence of mechanical isolation devices or direct barrier logs, a carefully sequenced pre-frac program was deployed to assess near-wellbore behavior, calibrate model assumptions, and set the stage for containment-focused stimulation.

Diagnostic Fracture Injection Test (DFIT)

The operation began with a Diagnostic Fracture Injection Test to assess the in-situ stress regime and validate the geomechanical profile. The DFIT included:

- Injection of a small volume of treated water at a low rate
- Observation of pressure decline after shut-in
- Identification of fracture closure pressure and leakoff behavior

The pressure falloff curve exhibited a distinct closure point, consistent with the modeled minimum horizontal stress in the 3D simulator. Importantly, no evidence of excessive leakoff or near-wellbore complexity was observed. The slope of the G-function and square root of time plots confirmed moderate formation toughness and supported the assumption of a single planar fracture.

This DFIT outcome provided two crucial assurances:

1. It validated the stress contrast below the perforated interval as minimal but stable.
2. It confirmed that the fracture growth profile assumed in design simulations could be trusted — particularly the projection of limited upward extension.

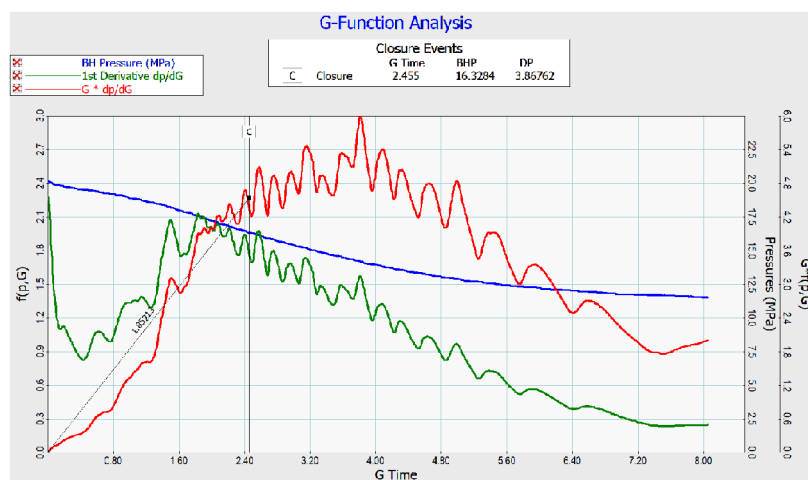


Figure 5—Breakdown test G-function study pointed to height recession. This likely to accord expected fracture geometry behavior.

Step-Down and Calibration Tests

Following the DFIT, a step-down test was performed to quantify near-wellbore friction and perforation pressure drop. This test yielded:

- A friction coefficient consistent with pre-job modeling
- No anomalous pressure spikes, indicating good cement or formation contact

This was followed by a calibration injection using the actual crosslinked fluid system, simulating early-stage main frac behavior. The observed breakdown pressure matched design expectations within 0.3 MPa, and the pressure curve showed no signature of height breakout during pad injection.

These tests collectively confirmed that fracture propagation was initiating in a contained fashion, without downward breach or unusual tortuosity.

Controlled Proppant Bridging Barrier Execution

Between the calibration and main fracture stages, a **controlled proppant bridging barrier** was deployed. This stage used a custom blend of intermediate and fine solids pumped at FET (using crosslinked fluid), designed to form a localized tip screenout or bridging effect near the lower extent of the perforated interval. The aim was to physically limit fracture penetration into the lower, low-stress claystone.

The bridging stage was pumped at a moderate rate ($\sim 2.0 \text{ m}^3/\text{min}$) and terminated upon reaching a surface pressure plateau, suggesting near-tip bridging. This operation functioned not only as a barrier but also as a pre-conditioning phase, raising near-wellbore resistance and modulating net pressure behavior in the following main frac.

Execution-phase diagnostics confirmed that the design's core assumptions — particularly around stress, growth direction, and pressure envelope — were valid in the field. By embedding diagnostics into the workflow and pairing them with a bridging strategy, the team was able to shift fracture behavior from reactive to controlled, dramatically improving the likelihood of a successful, confined treatment.

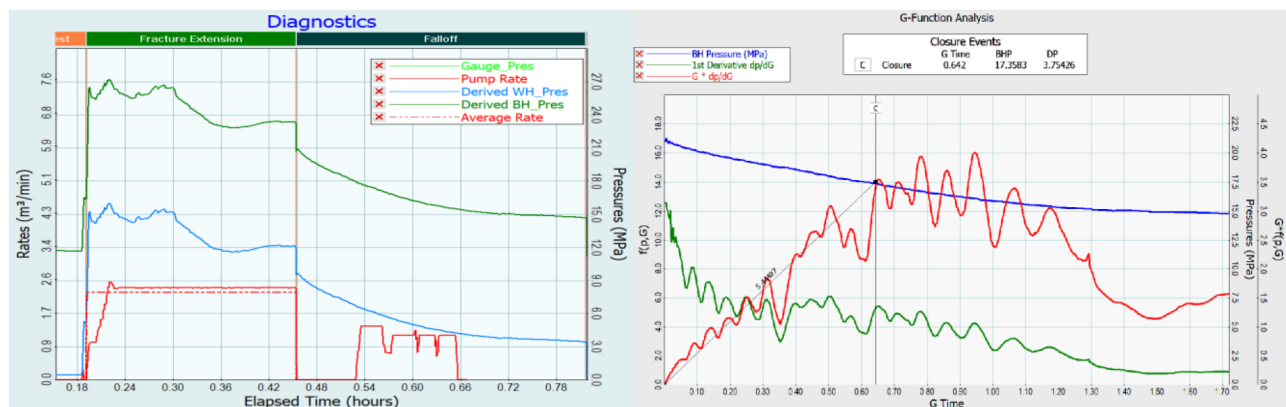


Figure 6—Fluid Efficiency Test with Controlled proppant bridging barrier pumped ahead of displacement

Main Fracturing Execution and Post-Job Evaluation

Following the placement of the controlled proppant bridging barrier — executed with linear gel and designed to condition the fracture tip — the main fracturing stage was initiated without operational delays. This transition marked the start of the primary stimulation phase intended to establish fracture conductivity within the 7-meter net pay zone.

Main Stage Pumping

The main treatment was pumped at a controlled rate of 2.25 m³/min, consistent with pre-job simulations, using a guar-based delayed crosslinked fluid. The proppant ramp consisted of:

- Intermediate Strength Proppant (ISP) in 30/50 and 20/40 mesh
- A gradual increase in proppant concentration up to 7 PPA
- A final tail-in using 16/30 ISP to strengthen near-wellbore conductivity

Throughout the pumping, surface pressure trends followed a consistent upward path, with no signs of abrupt pressure loss, instability, or rapid fluid loss indicative of vertical growth into low-stress formations. This smooth execution, coupled with pressure behavior consistent with model expectations, indicated that the prior bridging stage had effectively altered near-tip conditions to support vertical containment.

Net Pressure Trends and Height Control Indicators

One of the primary indicators of effective fracture containment was the net pressure behavior observed during the main stage. As the treatment progressed, surface pressure continued to rise — a response typically associated with increased near-tip resistance and limited vertical growth.

While pre-job simulations indicated a total fracture height of approximately 21 meters, this modeled envelope does not represent the full extent of conductive propped fracture. The observed net pressure increase, coupled with the absence of sudden pressure drops, suggested that conductive growth remained centered within the perforated interval, rather than extending into adjacent clay-rich formations.

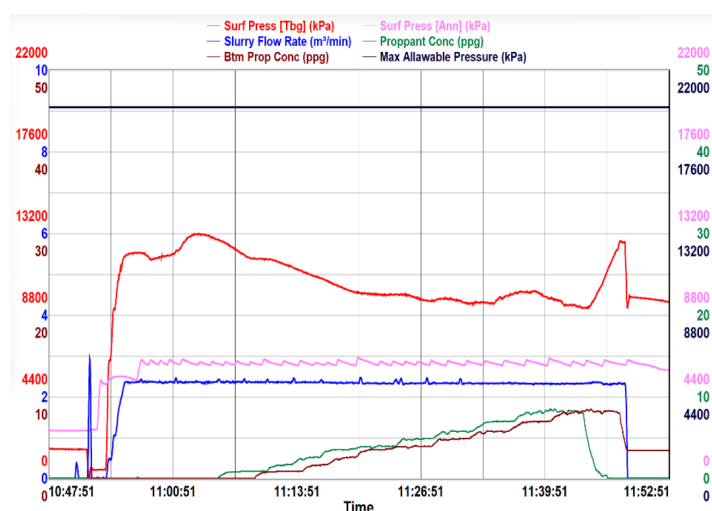


Figure 7—Main Treatment Plot. PAD stage part demonstrates net pressure positive trend.

Post-Job Diagnostics

Two forms of diagnostics were used to validate containment:

- **Temperature Monitoring:** Downhole gauges tracked thermal response post-frac, revealing temperature drops consistent with fluid placement across the perforated zone, with no anomalies indicating vertical breakout.
- **Tracer Evaluation:** Tracers embedded in each proppant stage were recovered selectively, with peak concentrations observed in zones aligned with the target interval. This supports the conclusion that fracture conductivity remained anchored within the intended reservoir layer.

These diagnostics, in combination with treatment pressures and lack of preliminary bridging or even pack-off or operational irregularity, provided strong evidence that the bridging barrier successfully modified tip conditions, enabling the main fracture to propagate in a vertically controlled manner.

Even in a thin, weakly confined reservoir, containment can be achieved without high-rate or high-viscosity designs. In this case, the deliberate placement of a bridging barrier using linear gel enabled controlled stimulation with crosslinked fluid — leading to a confined, productive fracture geometry supported by both execution trends and post-job diagnostics.

Discussion: Interpretation and Broader Implications

The Project Well case demonstrates that effective fracture containment in shallow, weakly bounded reservoirs can be achieved through targeted operational design — even in the absence of high stress contrast or robust natural barriers. While the fracture height in simulation extended beyond the perforated interval (~21 meters), both execution-phase pressure behavior and post-job diagnostics indicated that the conductive portion of the fracture remained confined to the 7-meter pay zone.

Design Philosophy Shift: Geometry Over Volume

Traditional fracturing in sandstone injectors often prioritizes proppant mass or fluid efficiency. In contrast, the Project Field design philosophy was centered on geometry control, driven by:

- Minimal mechanical contrast across layers
- Risk of vertical fracture migration reducing net pressure
- Strategic need to protect perforated zone conductivity

Rather than scaling proppant intensity or fluid complexity, the engineering team chose a low-rate, low-viscosity design paired with a bridging strategy — demonstrating that effective stimulation is not always volume-dependent.

Role of Preconditioning and Barrier Placement

The controlled proppant bridging barrier played a dual role:

- As a physical impedance, altering near-tip conditions and modifying stress gradients
- As a preconditioning step, improving pressure efficiency and guiding fracture propagation within target bounds

Its success underpins a broader implication: barrier placement can be a valid alternative to mechanical isolation or aggressive fluid rheology, especially when wellsite simplicity and operational cost control are priorities.

Fracture Height Interpretation: Simulation vs. Reality

The apparent contradiction between simulated height (~21 m) and confined conductive fracture is resolved by recognizing that numerical models predict total fracture envelope, including non-conductive, aperture-limited wings. In this case:

- The lower portion of the modeled height likely saw fracture propagation without proppant transport
- Near-tip resistance and fluid loss mechanisms contained proppant within a narrower, productive band
- Such distinctions highlight the importance of diagnostics and stress matching, not just modeling, when validating fracture geometry.

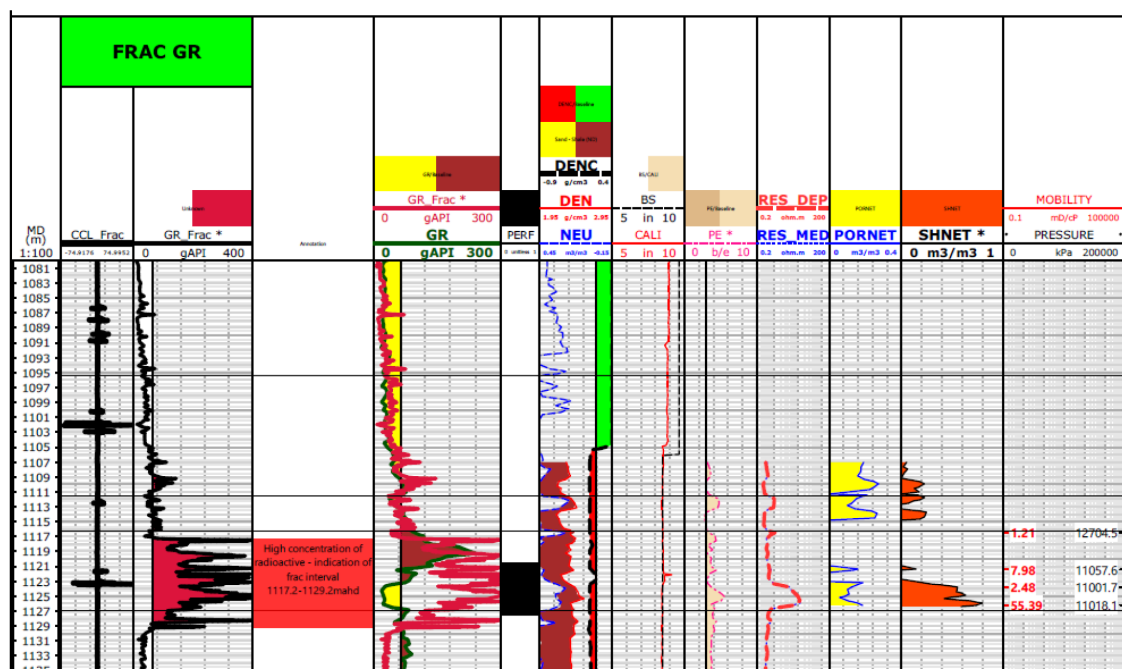


Figure 8—Main Treatment results reflection from Tracer logging

The Project well confirms that effective stimulation is not purely a function of stress contrast or mechanical barrier presence. By pairing geometry-driven modeling with execution-phase containment

mechanisms, the team demonstrated a viable method for frac design in thin, sensitive sandstone injectors. This approach offers a replicable framework for similar formations across PDO's portfolio and beyond.

Conclusion and Lessons Learned

The stimulation of the Project well in the western Project Field illustrates a pragmatic and effective approach to hydraulic fracturing in shallow, vertically sensitive sandstone reservoirs. Despite minimal stress contrast and the absence of mechanical barriers, the operation achieved a confined, productive fracture geometry by combining thoughtful modeling, diagnostics, and a targeted near-tip intervention.

The Jurassic Project Formation, although typically encountered at greater depths, was successfully stimulated at approximately 1100 mMD due to favorable uplift in the Block X western margin. Fracture height control emerged as the primary design concern, outweighing traditional considerations such as fluid rheology optimization or proppant mass maximization. A controlled proppant bridging barrier, executed using linear gel, proved instrumental in modulating near-tip behavior and elevating net pressure during the main stage. Although the 3D simulator predicted fracture heights of around 21 m, diagnostic observations showed that conductive height remained confined to the targeted perforated zone. Execution pressures and post-job tracer evaluation further confirmed vertical containment without reliance on mechanical isolation tools or complex fluid systems.

Several lessons can be drawn from this experience. Containment can be achieved with relative simplicity, as linear gel in a carefully placed barrier stage effectively modified fracture tip behavior and reduced growth into non-target intervals. Cross-validation of the stress model with DFIT results improved confidence in pre-frac assumptions and helped avoid surprises during execution. Diagnostics proved essential; temperature profiling and tracer recovery were critical in confirming that modeled behavior matched field reality. Moreover, barrier strategies should not be dismissed in thin-pay or weakly confined reservoirs, as they provide a cost-effective geometry control option, particularly when mechanical alternatives are impractical.

In an environment where traditional frac design is often centered on scale, this case underscores that geometry — not volume — defines success in thin reservoirs. The Project Well exemplifies how an intelligent, diagnostics-integrated design can deliver effective, contained stimulation even in shallow Jurassic settings.

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